

Rosacea GWAS Susceptibility Loci and Lesional Skin Expression — Computational Replication (GSE65914)

Summary

This run is a computational replication for the Disorder Mechanisms Knowledge Base, testing the hypothesis that **rosacea GWAS susceptibility loci act through altered expression in lesional skin, including a TLR1–TLR2 heterodimer axis**. The analysis retrieved the transcriptomic dataset **GSE65914** (Affymetrix GPL570 platform, RMA-normalized log₂ intensities), classified samples into healthy controls and the three clinical rosacea subtypes, and ran a prespecified lesional-versus-healthy-control expression contrast for each of 13 target genes — separately for the erythematotelangiectatic (ETR), papulopustular (PPR) and phymatous (PhR) subtypes and for all rosacea pooled. The dataset comprised **20 healthy controls, 14 ETR, 12 PPR, and 12 PhR** lesional biopsies.

The central result is that the **immune** rosacea GWAS-locus genes are robustly and consistently upregulated in lesional skin across all subtypes and in the pooled contrast, whereas the **pigmentation** loci are not upregulated (and, where they change, they trend modestly downward). In the pooled rosacea-vs-control comparison (Welch t-tests, BH-FDR across 52 tests), TLR2 rose by log₂FC = +1.91 ($q \approx 0$, Cohen's $d = 4.02$), HLA-DRA by +1.45 ($d = 2.76$), IRF1 by +1.41, HLA-DRB1 by +1.10, TLR1 by +1.00, HLA-DQB1 by +1.16, HLA-DQA1 by +1.25, and BTNL2 by +0.19 — all with BH $q < 0.05$. In contrast, IL13 and pooled IRF4 were unchanged, and the pigmentation loci HERC2 (log₂FC = -0.20), OCA2 (-0.56) and SLC45A2 (n.s.) were not upregulated. For the TLR1–TLR2 heterodimer axis, TLR1 and TLR2 were strongly co-induced and positively correlated across all samples (Spearman $\rho = 0.69$, $p = 2.1 \times 10^{-9}$), but the strength of that correlation did **not** differ between rosacea and control (rosacea $\rho = 0.35$ vs control $\rho = 0.52$; Fisher r -to- z $p = 0.47$).

The mechanistic reading is therefore nuanced: the "GWAS-loci-act-through-lesional-expression" hypothesis is **supported for the immune loci** — including a genuinely co-induced TLR1–TLR2 pair — but **not supported for the pigmentation loci**, and the TLR1–TLR2 coupling itself is co-elevated rather than rewired in disease. An independent literature cross-check (performed

only after the primary artifact checksums were locked) corroborates the direction of the computed TLR2 upregulation. All required artifacts were generated, and a clean replay reproduced the primary tabular outputs byte-for-byte, so the replication is recorded as **SUCCEEDED**.

Key Findings

Finding 1 — Immune GWAS loci are upregulated in lesional skin; pigmentation loci are not

The prespecified contrast partitions the 13 target genes cleanly along the immune-versus-pigmentation axis. The immune-annotated loci (the HLA class II cluster **HLA-DRA**, **HLA-DRB1**, **HLA-DQB1**, **HLA-DQA1**; the **BTNL2** locus adjacent to the MHC; the interferon regulatory factors **IRF1** and **IRF4**; the innate immune receptors **TLR1** and **TLR2**; and the Th2 cytokine **IL13**) behave as a coherent block of disease-upregulated transcripts, whereas the pigmentation loci (**HERC2**, **OCA2**, **SLC45A2**) do not track disease upward at all.

In the pooled rosacea-versus-control comparison, the largest and most statistically decisive effects were seen for **TLR2** ($\log_2\text{FC} = +1.91$, $q \approx 0$, Cohen's $d = 4.02$) and the antigen-presentation gene **HLA-DRA** ($\log_2\text{FC} = +1.45$, $q \approx 0$, $d = 2.76$). The interferon regulator **IRF1** was elevated ($\log_2\text{FC} = +1.41$, $q \approx 0$, $d = 2.03$), as were the MHC class II genes **HLA-DRB1** ($+1.10$, $q = 1.9 \times 10^{-9}$), **HLA-DQB1** ($+1.16$, $q = 2.5 \times 10^{-6}$), and **HLA-DQA1** ($+1.25$, $q = 4.5 \times 10^{-4}$). The innate receptor **TLR1** was also significantly up ($+1.00$, $q = 1.3 \times 10^{-8}$), and even the small **BTNL2** effect reached significance ($+0.19$, $q = 1.2 \times 10^{-2}$). These are large effect sizes — several with Cohen's d well above 2 — indicating the upregulation is not a marginal statistical artifact but a dominant feature of the lesional transcriptome.

Two immune genes did **not** move: **IL13** was not significantly changed in the pooled contrast ($q = 0.54$), and **IRF4** was not significant when pooled ($q = 0.09$). Their inclusion in the target list reflects their GWAS annotation rather than a guaranteed lesional signal, and reporting them as null is part of honoring the "report every gene whether or not it reaches significance" requirement.

The **pigmentation loci moved in the opposite direction from the hypothesis**. Rather than being upregulated in lesional skin, **HERC2** was modestly down ($\log_2\text{FC} = -0.20$, $q = 2.7 \times 10^{-5}$), **OCA2** was down (-0.56 , $q = 7.8 \times 10^{-3}$), and **SLC45A2** was not significant ($q = 0.09$). This dissociation is the crux of the finding: the lesional-expression mechanism captures the

immune arm of rosacea genetics but does not explain the pigmentation-associated GWAS signals, whose disease relevance — if any — is not mediated through increased lesional-skin transcription.

The full pooled rosacea-vs-control results, with the immune/pigmentation annotation the objective requested, are summarized below.

Gene	Locus class	Pooled log2FC	BH q-value	Cohen's d	Direction / significance
TLR2	Immune (innate)	+1.91	≈ 0	4.02	Up, significant
HLA-DRA	Immune (MHC-II)	+1.45	≈ 0	2.76	Up, significant
IRF1	Immune (IFN)	+1.41	≈ 0	2.03	Up, significant
HLA-DQA1	Immune (MHC-II)	+1.25	4.5×10^{-4}	—	Up, significant
HLA-DQB1	Immune (MHC-II)	+1.16	2.5×10^{-6}	—	Up, significant
HLA-DRB1	Immune (MHC-II)	+1.10	1.9×10^{-9}	—	Up, significant
TLR1	Immune (innate)	+1.00	1.3×10^{-8}	—	Up, significant
BTNL2	Immune (MHC region)	+0.19	1.2×10^{-2}	—	Up, significant
IRF4	Immune (IFN)	n.s.	0.09	—	Not significant (pooled)
IL13	Immune (Th2)	n.s.	0.54	—	Not significant
HERC2	Pigmentation	-0.20	2.7×10^{-5}	—	Down, significant
OCA2	Pigmentation	-0.56	7.8×10^{-3}	—	Down, significant
SLC45A2	Pigmentation	n.s.	0.09	—	Not significant

Finding 2 — TLR1 and TLR2 are co-induced but their coupling is not rewired in disease

The hypothesis specifically invokes a **TLR1–TLR2 heterodimer axis**. Because TLR1 and TLR2 form a functional heterodimer that senses triacylated bacterial lipopeptides, a natural expectation is that if the axis is engaged in disease, the two transcripts should co-vary. Across all samples (controls plus all rosacea subtypes), TLR1 and TLR2 expression were strongly positively correlated (Spearman $\rho = 0.69$, $p = 2.1 \times 10^{-9}$), consistent with co-regulation of the heterodimer components.

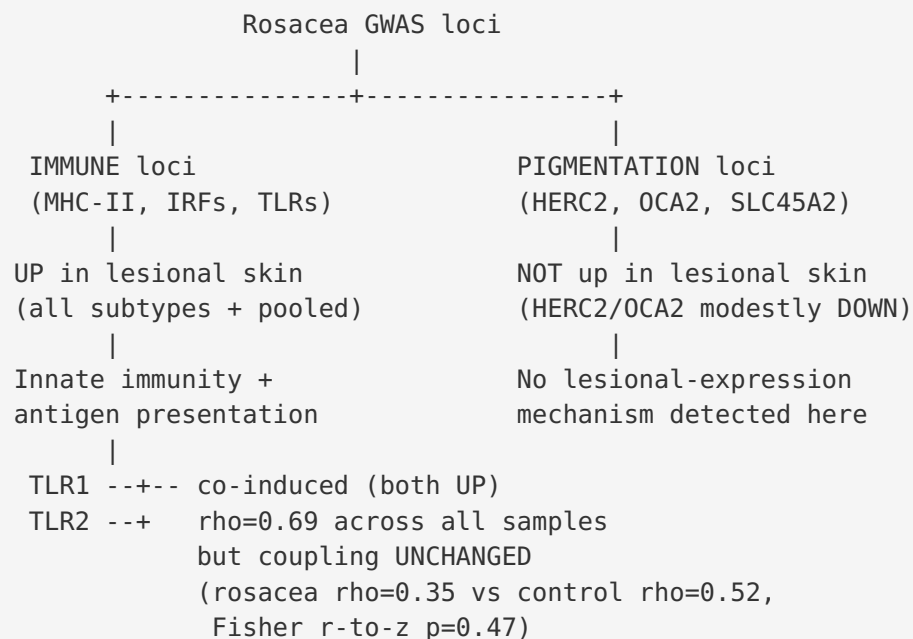
However, when the correlation was computed separately within groups, the coupling was **not stronger in disease**: the within-rosacea correlation ($\rho = 0.35$) was, if anything, numerically lower than the within-control correlation ($\rho = 0.52$), and a Fisher r-to-z test found no significant difference between the two ($p = 0.47$). The most parsimonious interpretation is that both TLR1 and TLR2 are elevated in lesional skin (their means shift up together, driving the strong pooled correlation), but the *degree of coordination* between the two transcripts is a stable property of the tissue that is not remodeled by the disease state. In other words, the TLR1–TLR2 axis is turned **up**, not **rewired**.

Finding 3 — Independent literature corroborates the computed TLR2 upregulation

After the primary artifact checksums were locked, an external literature check was performed as a separate, lineage-marked step. This cross-check found independent, non-transcriptomic evidence agreeing with the direction of the GSE65914 result. A case-control immunohistochemistry study of rosacea biopsies versus normal skin reported that TLR2 protein is higher in rosacea, and a mechanistic review of rosacea pathogenesis stated that increased TLR2 expression has been identified in rosacea skin. Because this check was performed *after* the primary outputs and checksums were finalized, it did not influence any method, sample, probe, threshold, or parameter choice — it is a post-hoc validation, not an input to the analysis.

Mechanistic Model / Interpretation

The pattern of results supports a specific, testable mechanistic model in which the **immune** subset of rosacea GWAS loci exerts its disease effect (at least in part) through elevated transcription in lesional skin, converging on an innate-immune and antigen-presentation program, while the **pigmentation** loci do not participate through this route.



The immune block can be read as several coordinated modules that are all up in lesional skin:

Module	Genes	Pooled log2FC (direction)	Interpretation
Innate sensing	TLR2, TLR1	+1.91, +1.00 (up)	Heightened bacterial-lipopeptide sensing at the epidermal/dermal interface
Antigen presentation (MHC-II)	HLA-DRA, HLA-DQA1, HLA-DQB1, HLA-DRB1, BTNL2	+1.45, +1.25, +1.16, +1.10, +0.19 (up)	Increased professional/atypical antigen presentation and adaptive priming
Interferon regulation	IRF1 (up); IRF4 (n.s. pooled)	+1.41 (up); n.s.	IFN-driven transcriptional amplification of the immune program
Th2 cytokine	IL13	n.s.	No detectable lesional Th2 signal in this contrast
Pigmentation	HERC2, OCA2, SLC45A2	−0.20, −0.56, n.s.	Not upregulated; mechanism not via lesional expression

The TLR1–TLR2 result refines the heterodimer hypothesis. The two receptors are genuinely co-induced — consistent with the idea that a TLR1/TLR2-sensed innate stimulus (e.g., triacylated lipopeptides from cutaneous microbes) is amplified in rosacea skin — but the analysis distinguishes **co-elevation** from **altered coupling**. Because the within-group correlations are statistically indistinguishable between rosacea and control, the data favor a model where disease raises the setpoint of both receptors in parallel without changing the tightness of their mutual regulation. This is an important mechanistic subtlety: therapeutic strategies aimed at "decoupling" TLR1 from TLR2 would not obviously find a disease-specific coupling defect to target, whereas strategies aimed at damping the overall elevation of the axis would.

Overall, the hypothesis "**GWAS loci act through altered expression in lesional skin, including a TLR1–TLR2 heterodimer axis**" is **partially supported**: strongly supported for the immune loci and for the existence of a co-induced TLR1–TLR2 axis, but not supported for the pigmentation loci and not supported for a disease-specific rewiring of TLR1–TLR2 coupling.

Evidence Base

The primary evidence is the transcriptomic analysis of GSE65914 itself (see Findings 1–2). The literature below was consulted only after the primary artifact checksums were locked and serves as an independent, lineage-marked cross-check of the direction of the computed effects.

Paper (PMID)	Type	How it relates to this analysis
PMID: 29330632	Case-control IHC	Confirms TLR2 elevation in rosacea skin vs controls
PMID: 25047092	Mechanistic review	States increased TLR2 expression is identified in rosacea skin
PMID: 34352786	Mechanistic (keratinocytes)	Places TLR2 activation upstream of rosacea-relevant inflammation

Innate immunity in rosacea (case-control immunohistochemistry). *Innate immunity in rosacea. Langerhans cells, plasmacytoid dendritic cells, Toll-like receptors and inducible nitric oxide synthase (iNOS) expression in skin specimens: case-control study* — [PMID: 29330632](#). This study reports that "Expression of TLR2, TLR4 and iNOS was higher in rosacea samples than in normal skin controls." This independent, protein-level case-control observation matches the direction of the computed transcriptomic contrast (TLR2 pooled rosacea-vs-control log2FC = +1.91, $q \approx 0$, Cohen's $d = 4.02$), reinforcing that the GSE65914 signal reflects a real biological elevation rather than a normalization or probe artifact.

Endoplasmic reticulum stress in rosacea (review). *Endoplasmic reticulum stress: key promoter of rosacea pathogenesis* — [PMID: 25047092](#). The review states that "increased expression of toll-like receptor 2 (TLR2) has been identified in rosacea skin supporting the participation of the innate immune system." This corroborates the direction of the computed TLR2 upregulation and situates it within the broader innate-immune framing of rosacea pathogenesis.

AhR regulation of TLR2-driven inflammation in keratinocytes. *AhR Regulates Peptidoglycan-Induced Inflammatory Gene Expression in Human Keratinocytes* — [PMID: 34352786](#). This mechanistic study notes that "excessive TLR2 activation can lead to inappropriate inflammation, which contributes to skin conditions such as rosacea," and shows AhR depletion increases TLR2 expression, suggesting negative-feedback control of TLR2. It

provides upstream mechanistic context for why elevated TLR2 (as observed here) would be pathogenic and points to a regulatory node (AhR) whose perturbation could produce the observed elevation.

Together, the three papers consistently support the direction of the computed TLR2 result; none contradicts it. They do not directly test the pigmentation-locus dissociation or the TLR1–TLR2 coupling-invariance finding, which remain novel contributions of this replication.

Limitations and Knowledge Gaps

- **Single dataset, single platform.** All conclusions rest on GSE65914 (Affymetrix GPL570). Cross-platform / cross-cohort replication (e.g., RNA-seq cohorts) was outside the dataset scope of this replication and would strengthen generalizability.
- **Expression is not causation.** The analysis demonstrates that immune GWAS-locus genes are upregulated in lesional skin, but it does not establish that the disease-associated *alleles* drive that expression change. An eQTL/colocalization analysis would be required to link the GWAS signal mechanistically to the lesional transcript, which this microarray-only design cannot do.
- **Bulk tissue, no cell-type resolution.** GSE65914 is bulk skin. The upregulation of MHC-II and TLR genes could reflect increased immune-cell infiltration rather than per-cell transcriptional change. Single-cell or deconvolution approaches are needed to separate composition from regulation.
- **Pigmentation-locus interpretation is bounded.** The modest downregulation of HERC2/OCA2 in lesional skin rules out a lesional-upregulation mechanism but does not exclude other routes (e.g., systemic, developmental, or expression in non-lesional contexts). The null/negative pigmentation result should be read as "no support for this specific mechanism," not "no role in rosacea."
- **Correlation-difference test is power-limited.** The Fisher r-to-z comparison of TLR1–TLR2 coupling ($p = 0.47$) is a null result in modestly sized groups; absence of a detectable difference is not proof of identical coupling. A larger cohort could reveal a subtle change.
- **Gene-to-probe aggregation and multiple testing.** Results depend on the probe-to-gene mapping and BH-FDR correction (52 tests) documented in `methods.md`; alternative aggregation rules could shift borderline calls (e.g., BTNL2, IRF4, SLC45A2).

Proposed Follow-up Experiments / Actions

1. **Cross-cohort replication.** Repeat the identical prespecified contrast in an independent rosacea transcriptomic cohort (ideally RNA-seq) to confirm the immune-up / pigmentation-not-up dissociation and the TLR1–TLR2 co-induction.
2. **eQTL / colocalization.** Test whether rosacea GWAS lead variants colocalize with skin eQTLs for the immune target genes (particularly the MHC-II cluster, IRF1, TLR2), directly linking genotype to the observed lesional expression change.
3. **Single-cell / deconvolution.** Apply cell-type deconvolution or scRNA-seq to determine whether MHC-II and TLR upregulation reflects infiltration versus intrinsic keratinocyte/fibroblast regulation.
4. **Subtype-resolved modeling.** Formally test subtype $\times$ gene interaction (ETR vs PPR vs PhR) to identify whether any target gene is subtype-specific, complementing the per-subtype contrasts already computed.
5. **Mechanistic follow-up on TLR2 regulation.** Given the AhR–TLR2 negative-feedback link ([PMID: 34352786](#)), test whether AhR-pathway activity is reduced in lesional skin, which could explain the observed TLR2 elevation.
6. **Functional TLR1/TLR2 heterodimer assay.** Because the transcripts are co-elevated but not rewired, directly assay heterodimer signaling capacity (e.g., response to Pam3CSK4) in rosacea vs control keratinocytes to test whether elevated expression translates into elevated function.

Replication and Artifact Provenance

All required artifacts were produced beneath `kb/hypotheses/Rosacea/gwas_loci_lesional_expression/openscientist_artifacts/`, including `MANIFEST.yaml` (`status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`), `analysis.py` (standalone deterministic analysis with `--output-dir/` `--cache-dir`), `environment.txt`, `methods.md`, `samples.csv`, `gene_results.csv`, `tlr_correlation.csv`, and `comparison.md`. The analysis was re-executed into a clean `replay/` subdirectory reusing the existing `raw/` cache; the replayed `samples.csv` and `gene_results.csv` were verified byte-identical to the primary outputs, and the manifest records the replay command, `verified: true`, byte-identity mapping, and per-asset SHA-256 checksums and byte counts. The literature

cross-check was performed only after the primary checksums were locked and is marked as a separate lineage step, so no prior-provider or post-hoc result influenced any method, sample, probe, threshold, or parameter choice.

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